A PROJECT TITLE

ON

NETVISION: High-Dimensional Network Log Visualisation Using PCA, t-SNE and Isolation Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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1. Abstract:

Modern networks generate large volumes of connection records described by dozens of numeric and categorical attributes. This makes manual inspection slow and hides patterns in raw tables. NETVISION is a machine-learning project that makes such high-dimensional network logs easier to explore. It combines data preprocessing, dimensionality reduction, visualisation and unsupervised anomaly detection in a single workflow. The project uses the UNSW-NB15 training dataset, which contains 175,341 records and 45 original columns. Exploratory analysis confirmed that the loaded data had no missing values and no duplicate rows. Categorical attributes such as protocol, service and state were encoded, expanding the feature space to 191 columns. The features were then standardised so that attributes with different units and ranges contribute comparably. Principal Component Analysis (PCA) reduced the encoded data from 191 features to 30 components, giving a compact representation of the traffic. To inspect local structure, t-SNE was applied to a sample of 2,000 records and projected into two dimensions. The resulting plots compare normal and attack-labelled traffic. Isolation Forest, an unsupervised anomaly detector, was then used to flag records whose feature patterns differ from the majority. With a 5% contamination setting, it flagged 8,767 records as potential anomalies and treated 166,574 as normal. A Streamlit dashboard presents the summaries and plots, and the results are exported as an anomaly CSV file and plots. The approach is primarily unsupervised. Dataset labels are used only to colour and compare plots, so no accuracy, precision, recall or F1-score is claimed. The flagged records are candidates for further investigation, not confirmed attacks, and the t-SNE view covers only a sample of the data. NETVISION therefore serves as an interpretable starting point for network-security education, exploratory analysis and analyst triage.

2. INTRODUCTION:

Network logs contain many features that are difficult to inspect directly, and high-dimensional data can hide patterns that matter for security. This project addresses that by combining PCA, t-SNE and Isolation Forest on the UNSW-NB15 intrusion dataset. The aim is to make the data easier to visualise and to highlight unusual records for review, without replacing security analysts. Exploratory data analysis guides the preprocessing. PCA gives a compact representation, t-SNE gives a two-dimensional view, and Isolation Forest flags potential anomalies. Because the pipeline is unsupervised, its output supports investigation rather than definitive attack decisions.

3. SOFTWARE REQUIREMENTS:

- **Language:** Python
- **Data handling:** Pandas, NumPy
- **Machine learning:** Scikit-learn (StandardScaler, PCA, t-SNE, Isolation Forest)
- **Visualisation:** Matplotlib, Plotly
- **Dashboard:** Streamlit
- **Dataset:** UNSW-NB15 training set (CSV)
- **Environment:** offline workflow; no database, special hardware or cloud services required

Signature of the Guide **Course Instructor**